Vienna Circle and Berlin Circle both formed in the 1920s, the same decade Heisenberg and Schrödinger (who were German and Austrian, respectively) first developed full-blown theories of quantum physics. This is not a coincidence, but it's not a conspiracy either. There were hazy ideas floating around the intellectual culture in their shared time and place that may have contributed to the ideas of both the early positivists and the first quantum physicists. But there were definitely specific common inspirations for both groups—most importantly, the work of Ernst Mach.

Mach, who worked at the University of Vienna a generation before the Vienna Circle, demanded that all scientific theories refer only to observable entities. (We first encountered him in Chapter 2; he denied the existence of atoms because they couldn't be seen, much to Ludwig Boltzmann's dismay.) Mach's observables-only philosophy of science was a direct inspiration for the development of logical positivism. The Vienna Circle even mentions him by name in its manifesto as one of its direct forerunners and most important influences. But Schlick, Neurath, and the rest weren't the only ones inspired by Mach. Mach was also godfather to a young Viennese mathematical prodigy by the name of Wolfgang Pauli. Mach's views permeated Pauli's philosophy of science. "There is no point in discussing…quantities [that] cannot, in principle, be observed experimentally," a young Pauli wrote, fresh out of college, in 1921. Such quantities, he maintained, would be "fictitious and without physical meaning." Over thirty years later, Pauli dismissed Einstein's concerns about the ability of quantum physics to describe what happens between measurements, saying that "one should no more rack one's brain about the problem of whether something one cannot know anything about exists all the same, than about the ancient question of how many angels are able to sit on the point of a needle."

Mach was not the only shared source of ideas for both the Vienna Circle and the physicists of Copenhagen. There was someone else both groups held in similar regard: Einstein, who had himself taken some inspiration from Mach in formulating special relativity. By holding fast to what could be observed—clocks and meter sticks—and dismissing the luminiferous aether as an unobservable phantom, Einstein revolutionized science, and the success of special relativity was seen as a triumph for the Machian view of physics. Certainly Heisenberg thought of relativity this way, as he told